3D object recognition

1. Content Description

This course implements the functions of acquiring color images and using the MediaPipe framework to recognize 3D objects (the examples only show four: a mug, a shoe, a chair, and a camera). This section requires entering commands in the terminal. The terminal to open depends on the motherboard type. This section uses a Raspberry Pi 5 as an example.

For Raspberry Pi and Jetson Nano boards, you need to open a terminal on the host computer and enter the command to enter the Docker container. Once inside the Docker container, enter the commands mentioned in this section in the terminal. For instructions on entering the Docker container from the host computer, refer to **[01. Robot Configuration and Operation Guide] -- [5.Enter Docker (For JETSON Nano and RPi 5)]**.

For Orin and RDK motherboards, simply open the terminal and enter the commands mentioned in this section.

2. Program Startup

For the Raspberry Pi 5 and jetson nano controller, you must first enter the Docker container. For the Orin and RDK controller, this is not necessary.

Entering the Docker container (see the Docker course for steps: 4. Docker Startup Script).

All commands below must be executed within the same Docker container **(See the Docker course for steps: 3. Docker Commit and Multi-Terminal Entry).**

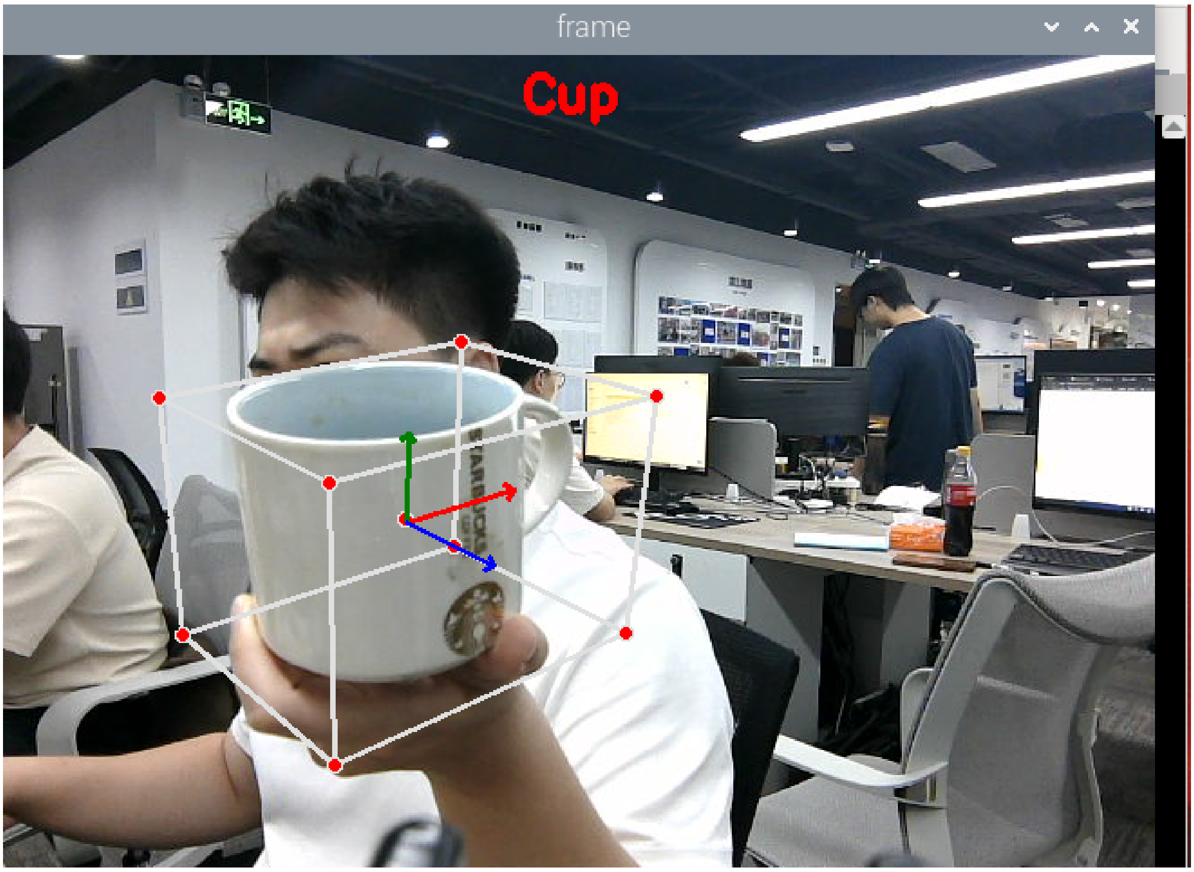
First, in the terminal, enter the following command to start the camera.

```
#usb camera
ros2 launch usb_cam camera.launch.py
#nuwa camera
ros2 launch ascamera hp60c.launch.py
```

After successfully starting the camera, open another terminal and enter the following command to start the 3D object detection program.

```
ros2 run yahboomcar_mediapipe 07_Objectron
```

After running the program, the default object to be recognized is shoes. You can press the F key to switch the object to be recognized. As shown in the image below, a mug is recognized.



3. Core Code Analysis

Program Code Path:

- Raspberry Pi 5 and Jetson Nano motherboard

The program code is running in Docker. The path in Docker is `/root/yahboomcar_ros2_ws/yahboomcar_ws/src/yahboomcar_mediapipe/yahboomcar_mediapipe/07_Objectron.py`

- Orin motherboard

The program code path is `/home/jetson/yahboomcar_ros2_ws/yahboomcar_ws/src/yahboomcar_mediapipe/yahboomcar_mediapipe/07_Objectron.py`

- RDK motherboard

The program code path is `/home/sunrise/yahboomcar_ros2_ws/yahboomcar_ws/src/yahboomcar_mediapipe/yahboomcar_mediapipe/07_Objectron.py`

Import the necessary library files.

```
import mediapipe as mp
import cv2 as cv
import time
import rclpy
from rclpy.node import Node
from sensor_msgs.msg import Image
from cv_bridge import CvBridge
import os
import numpy as np
```

Initialize data and define publishers and subscribers,

```
def __init__(self, staticMode=False, maxObjects=5, minDetectionCon=0.5,
minTrackingCon=0.99):
    super().__init__('objectron')
    self.staticMode = staticMode
    self.maxObjects = maxObjects
    self.minDetectionCon = minDetectionCon
    self.minTrackingCon = minTrackingCon
    self.index = 0
    self.modelNames = ['Shoe', 'Chair', 'Cup', 'Camera']
    #Use the class in the mediapipe library to define a 3D detection object
    self.mpObjectron = mp.solutions.objectron
    self.mpDraw = mp.solutions.drawing_utils
    self.mpobjectron = self.mpObjectron.Objectron(
    self.staticMode, self.maxObjects, self.minDetectionCon, self.minTrackingCon,
self.modelNames[self.index])
    self.bridge = CvBridge()
    #Define subscribers for the color image topic
    camera_type = os.getenv('CAMERA_TYPE', 'usb')
    topic_name = '/ascamera_hp60c/camera_publisher/rgb0/image' if camera_type ==
'nuwa' else '/usb_cam/image_raw'
    self.subscription = self.create_subscription(
    Image,
    topic_name,
    self.image_callback,
    10)
```

Color image callback function,

```
def image_callback(self, msg):
    frame = self.bridge.imgmsg_to_cv2(msg, desired_encoding='bgr8')
    action = cv2.waitKey(1)
    #Press the F key to switch the recognized objects
    if action == ord('f') or action == ord('F') : self.configUP()
    frame = self.findObjectron(frame)
    cv.imshow('Objectron', frame)
```

configUP switches the object recognition function and selects self.modelNames by modifying the value of self.index. The options for self.modelNames are ['Shoe', 'Chair', 'Cup', 'Camera']

```
def configUP(self):
    self.index += 1
    if self.index>=4:self.index=0
        self.mpobjectron = self.mpObjectron.Objectron(
        self.staticMode, self.maxObjects, self.minDetectionCon,
self.minTrackingCon,            self.modelNames[self.index])
```

findObjectron object recognition function,

```
def findObjectron(self, frame):
    cv.putText(frame, self.modelNames[self.index], (int(frame.shape[1] / 2) -
30, 30),
    cv.FONT_HERSHEY_SIMPLEX, 0.9, (0, 0, 255), 3)
```

```
    #Convert the color space of the incoming image from BGR to RGB to facilitate
subsequent image processing
    img_RGB = cv.cvtColor(frame, cv.COLOR_BGR2RGB)
    #Call the process function in the mediapipe library to process the image.
During init, the self.pose object is created and initialized.
    results = self.mpobjectron.process(img_RGB)
    #Determine whether the target object is recognized
    if results.detected_objects:
        for id, detection in enumerate(results.detected_objects):
            #On the image, draw the coordinate system for the identified target
object
            self.mpDraw.draw_landmarks(frame, detection.landmarks_2d,
self.mpObjectron.BOX_CONNECTIONS)
            self.mpDraw.draw_axis(frame, detection.rotation,
detection.translation)
    return frame
```